



Models/Analyses for IBL Dataset

Model / Analysis	Goal	What it tells you	Pros	Cons / Caveats	Feasibility (6 mo ISP)
PSTH + Modulation Index (MI)	Baseline stimulus encoding	Which neurons/regions respond differently to left vs right	Simple, interpretable; strong baseline	Doesn't separate stimulus vs choice vs movement	✓ Very feasible
Change-point Latency Estimation	Temporal hierarchy	When neurons/regions first encode laterality	Direct onset timing; aligns with prior IBL work	Sensitive to noise, smoothing choice matters	✓ Feasible
Demixed PCA (dPCA)	Dimensionality reduction	Separates population activity into stimulus, choice, time components	Visualization-friendly; widely used	Linear, may miss nonlinear codes	✓ Feasible
Targeted Dimensionality Reduction (TDR)	Axis-based decoding	Trajectories along stimulus vs choice axes	Clean trajectories, intuitive plots	Axes depend on trial averages; no unique variance	✓ Feasible

Mutual Information (MI) & Conditional MI	Information-theoretic selectivity	Quantify unique coding of stimulus vs choice	Nonparametric; doesn't assume linearity	Needs large samples; estimators biased in small bins	⚠ Moderate (careful estimation needed)
Partial Information Decomposition (PID)	Info decomposition	Unique vs redundant coding of stimulus/choice	Clean separation of overlaps	Computationally heavy; cutting edge	⚠ Moderate stretch goal
Cross-Condition Generalization (CCGP)	Identification test	Whether choice code generalizes beyond stimulus confounds	Strong identification; intuitive	Needs carefully defined splits; data-hungry	✅ Feasible if scoped
Gaussian-Process Factor Analysis (GPFA)	Latent trajectories	Smooth trial-level low-d dynamics; early vs late codes	Captures continuous latent dynamics	Computationally heavy on full IBL scale	⚠ Feasible with downsampling
Switching LDS / ARHMM	Discrete latent states	Identify “sensory → decision → movement” states	Matches decision stages; interpretable	Model complexity; hard EM fits	⚠ Stretch goal
Preferential Subspace ID (PSID)	Behavior-linked dynamics	Separate subspace predictive of choice vs sensory	Modern; aligns with movement-free codes	Newer, less standardized	⚠ Stretch goal

Linear Decoder (SVM/LogReg)	Predict choice/stimulus	Performance baseline for decoding	Fast, interpretable weights	Linear only; may underfit	✓ Easy baseline
Temporal CNN (TCN) / RNN	Nonlinear temporal decoding	Choice prediction from population spiking over time	Strong performance; captures dynamics	Risk of overfitting/leakage; needs careful CV	⚠ Moderate if careful
Tree Ensembles (XGBoost)	Interpretable ML decoding	Predict choice, rank feature importance	Handles nonlinearities; SHAP values explainable	Needs feature engineering; weaker temporal modeling	✓ Feasible
Virtual Lesions (ablation in decoders)	Regional contribution	How decoding drops if you remove region(s)	Maps functional importance directly	Depends on decoder stability	✓ Straightforward extension
Natural experiment (block switches)	Probe prior expectation encoding	Whether code shifts at prior block boundaries	Identifies prior-related signals	Sparse trials per block switch; noisy	⚠ Feasible with aggregation
Zero-contrast trial analysis	Instrumental variable for choice	Identify choice signals independent of stimulus	Cleanest separation of choice	Limited trial counts; weaker SNR	✓ Feasible, good control